Andrea Panebianco

Postdoctoral Researcher

Department of Electrical and Computer Engineering, Auburn University
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Google Scholar: h-index 8, 152 citations · Scopus: 128 citations · as of October 2026

RESEARCH INTERESTS AND PROFILE

Interests: goal-oriented and semantic communications, including Variational Information Bottleneck (VIB) methods; multi-agent and distributed online learning (multi-armed and restless bandits, reinforcement learning) for network control; cross-layer optimization of multi-hop wireless networks; Age of Information (AoI) and timeliness; stochastic channel modelling and prediction; Internet of Underwater Things (IoUT); NextG and Software-Defined Radio (SDR) testbeds for mission-critical IoT; computer vision for safety monitoring.

Profile. Researcher in wireless networking focused on goal-oriented and semantic communications, where information is transmitted according to its significance for the task at the receiver. The work combines learning-based and model-based decision making for resource-constrained networks: fully distributed (multi-agent) and hierarchical multi-armed bandits, reinforcement learning, and distributed convex optimization based on the proximal Alternating Direction Method of Multipliers (ADMM), together with stochastic channel models (Markov chains, hidden Markov models, Kalman filtering). These tools are applied to link adaptation (modulation, power, OFDM carrier selection), feedback scheduling, routing, and congestion control, with timeliness quantified by the AoI. The methods were first developed for underwater acoustic networks, with channel models derived from real experimental traces and protocols evaluated in network simulation, and are now being extended to NextG wireless testbeds and edge-intelligence platforms for semantic communication and disaster response. Additional experience includes research proposal preparation, conference organization, and Ph.D. mentoring.

EDUCATION

Ph.D. in Information and Communication Technologies Nov. 2022 – Dec. 2025
Department of Engineering, University of Palermo, Palermo, Italy
Degree awarded: 17 Dec. 2025
Thesis: "Learning and Adaptation for Efficient and Resilient Internet of Underwater Things Communications"
Supervisor: Prof. Sergio Palazzo
Three-part thesis on IoUT communications: (i) model-based channel prediction from real acoustic traces; (ii) lightweight multi-armed bandit frameworks for physical-layer-aware link adaptation (modulation and power control), distributed (multi-player) decision making, and AoI-aware feedback scheduling, targeting energy-efficient resource allocation; (iii) bandit- and reinforcement-learning-based routing for energy balance and jamming resilience. Validated through network simulations under realistic channel, latency, and energy constraints.

M.Sc. in Telecommunications Engineering Oct. 2019 – Oct. 2022
Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
Degree awarded: 26 Oct. 2022
Thesis: "Characterization of Underwater Communication Channels in Shallow Water Environments: From On-Field Measurements to Markov-Based Models"
Supervisor: Prof. Sergio Palazzo
Co-supervisors: Prof. Laura Galluccio, Dr. Fabio Busacca
Finite-state Markov models of shallow-water acoustic channels, derived from and validated against experimental traces of the LOON testbed of the NATO Centre for Maritime Research and Experimentation (CMRE): acoustic trace processing, channel impulse response analysis, SNR estimation, channel-quality assessment, propagation-induced variability, and MATLAB-based comparison of candidate models.

B.Sc. in Electronic Engineering Sep. 2015 – Mar. 2020
Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
Degree awarded: 20 Mar. 2020
Thesis: "Implementation of a Haptic Interface for Tactile Internet Scenarios in 5G Networks"
Supervisor: Prof. Sergio Palazzo
Co-supervisor: Prof. Giovanni Schembra
Hapkit-based haptic interface with Arduino device control and a Mininet/SDN emulation environment; packet-level latency evaluation under different network-delay and routing conditions.

RESEARCH EXPERIENCE

Postdoctoral Researcher Jan. 2026 – Present
Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA
Supervisor: Prof. Yin Sun
Research on adaptive communication systems for resource-constrained wireless networks, semantic and goal-oriented communications, and learning-based network control (cross-layer optimization, online learning, routing, scheduling, and resource allocation in dynamic environments) within externally funded projects (see *Research Projects*); contributions to research proposals; mentoring of Ph.D. and undergraduate students.

Visiting Ph.D. Student Jan. 2025 – Jul. 2025
Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA
Host: Prof. Yin Sun
Learning-based and AoI-aware communication mechanisms for underwater acoustic networks under limited feedback: adaptive modulation and power selection, feedback scheduling, and bandit-based resource allocation, aimed at lightweight online algorithms that improve reliability, energy efficiency, and adaptability (work supported in part by the U.S. National Science Foundation CAREER award CNS-2239677). Outputs: [C4, PA2][1].

Research Affiliate Nov. 2022 – Dec. 2025
Catania Research Unit, National Inter-University Consortium for Telecommunications (CNIT), Catania, Italy
Affiliation held throughout the Ph.D. (see *Research Projects*).

Research Intern Sep. 2020 – May 2021
Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
Research and development on underwater communication systems and IoT-based monitoring architectures within the project "Secure and Efficient Wireless Communications" (see *Research Projects*).

RESEARCH CONTRIBUTIONS

- **Goal-oriented and semantic communications.** Significance-driven, spectrum-efficient semantic transmission for mission-critical monitoring, the subject of two U.S. provisional patent applications [PA1, PA2]. SafeStep and MetaVIB-Air, a browser-based platform and an SDR testbed that demonstrate semantic communication for pedestrian safety monitoring by combining computer vision on live traffic-camera feeds with a VIB-based semantic transceiver, developed as co-first author [C2, P2].
- **Distributed online learning for network adaptation and timeliness.** Fully distributed learning frameworks in which each node decides from local feedback: multi-player bandits for modulation adaptation [J1, C7], a bilevel bandit that jointly adapts modulation, power, and the AoI-aware feedback interval under limited feedback [C4], and energy-fair bandit-based routing for multi-hop LoRa networks [C8]. Complementary lightweight bandit frameworks address multi-objective modulation adaptation, power control, OFDM carrier selection, and network cost-effectiveness on resource-constrained underwater nodes [J2, C5, C6, C9, C10], and learning-based routing provides resilience to jamming [J5]. Two extensions are under review: the bilevel bandit is reformulated as a partially observable Markov decision process with decision-dependent feedback delay and layerwise regret guarantees [U1], and OFDM carrier selection is generalized to joint subcarrier and power allocation through nested potential games [U2]. A survey of bandit methods for resource-constrained wireless devices, with a quantitative criterion of effective statelessness to decide when bandits can replace stateful reinforcement learning, is under review [U3].
- **Cross-layer optimization under covertness and mobility.** The first distributed cross-layer algorithm for joint congestion control, routing, scheduling, and power control in covert multi-hop, multi-modal networks, built on a concave bound on the detection error probability and on a parallel proximal ADMM with proven global linear convergence and robust tracking under fading and warden mobility, developed as co-first author [C1, P1]. A velocity-informed scheme adapts transmission parameters in high-mobility IoT scenarios [C3].
- **Stochastic channel modelling and prediction.** Finite-state Markov and hidden Markov channel models and predictive estimators, including Kalman filtering, derived from real shallow-water acoustic traces of the NATO CMRE LOON testbed [J4, C11, P3], and a comparative study of predictive versus adaptive techniques in terms of accuracy and complexity [J3].
- **Experimental IoUT systems.** An integrated acoustic/LoRa system for the transmission of multimedia sensor data over an IoUT, validated in nearshore shallow-water field trials [J6].

[1] Bracketed labels refer to the numbered lists in *Publications* and *Patent Applications*: [J] journal articles, [C] conference papers, [P] posters and demonstrations, [U] manuscripts under review, [PA] patent applications.

HONOURS AND AWARDS

Selected Best Paper 2024
IEEE International Conference on Communications (ICC 2024), Denver, CO, USA
For "Adaptive Modulation in Underwater Acoustic Networks (AMUSE): A Multi-Armed Bandit Approach" [C9] (extended version: [J2]).

Francesco Carassa Award (Best Poster) Sep. 2023
Annual Meeting of the Italian National Group on Telecommunications and Information Theory (GTTI 2023), Rome, Italy
For "Characterization of Underwater Acoustic Channel Models for SNR Prediction in Real Shallow Water Environment" [P3].

Student Award "TLC for a Better World" (Innovative Prototype Category) Dec. 2019
TelcomDay 2019, Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
For "Implementation of a Haptic Interface for Tactile Internet Scenarios in 5G Networks".

PUBLICATIONS

6 peer-reviewed journal articles · **11** peer-reviewed conference papers · **3** posters and demonstrations · **3** manuscripts under review (not counted above). Entries are listed in reverse chronological order within each category.

Notation. **Panebianco, A.** denotes the author of this CV. Underlined co-authors are students mentored or co-supervised by the author (Ph.D., M.Sc., or undergraduate). The mark † denotes equal contribution (co-first authors). The mark $^{\alpha}$ denotes an author list in alphabetical order, the convention of the authors' research group; in all other entries, the order reflects contribution.

Peer-Reviewed Journal Articles

[J1] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R.$^{\alpha}$ (2026). Bandits Under the Waves: A Fully-Distributed Multi-Armed Bandit Framework for Modulation Adaptation in the Internet of Underwater Things. *IEEE Transactions on Network and Service Management*, 23, 639–653.

[J2] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R.$^{\alpha}$ (2025). AMUSE: A Multi-Armed Bandit Framework for Energy-Efficient Modulation Adaptation in Underwater Acoustic Networks. *IEEE Open Journal of the Communications Society*, 6, 2766–2779. *Extended version of [C9].*

[J3] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, Qi, Z., & Pompili, D. (2024). Adaptive versus Predictive Techniques in Underwater Acoustic Communication Networks. *Computer Networks*, 252, 110679. *Authors 1–5 in alphabetical order.*

[J4] Busacca, F., Galluccio, L., Palazzo, S., & **Panebianco, A.**$^{\alpha}$ (2024). A Comparative Analysis of Predictive Channel Models for Real Shallow-Water Environments. *Computer Networks*, 250, 110557. *Based on experimental acoustic traces provided by NATO CMRE.*

[J5] Mertens, J. S., **Panebianco, A.**, Surudhi, A., Prabagarane, N., & Galluccio, L. (2023). Network Intelligence vs. Jamming in Underwater Networks: How Learning Can Cope with Misbehavior. *Frontiers in Communications and Networks*, 4, 1179626.

[J6] Brincat, A. A., Busacca, F., Galluccio, L., Mertens, J. S., Musumeci, A., Palazzo, S., & **Panebianco, A.**$^{\alpha}$ (2022). An Integrated Acoustic/LoRa System for Transmission of Multimedia Sensor Data over an Internet of Underwater Things. *Computer Communications*, 192, 132–142.

Peer-Reviewed Conference Papers

[C1] Chakraborty, S.†, **Panebianco, A.**†, Tian, Y., Chan, K. S., Dagefu, F., Sun, Y., & Shroff, N. B. (2026). Distributed Cross-Layer Optimization for Covert Multi-Hop, Multi-Modal Networks: Exponentially Fast Convergence and Robust Tracking. In *Proc. IEEE Military Communications Conference (MILCOM 2026)*, Oct. 2026. *Accepted; preprint arXiv:2608.28469.*

[C2] McDowell, C.†, **Panebianco, A.**†, Yang, J.†, Chakraborty, S., Chamoun, S., Ross, T., & Sun, Y. (2026). SafeStep: An Interactive Demonstration of Semantic Communication for Pedestrian Safety Monitoring. In *Proc. IEEE Military Communications Conference (MILCOM 2026), Workshop on Quality, Value, and Age of Information for Tactical Networks (WS7)*, Oct. 2026. *Accepted; preprint arXiv:2608.27688; live system: safestep.eng.auburn.edu.*

[C3] Busacca, F., **Panebianco, A.**, Raftopoulos, R., & Scuderi, D.$^{\alpha}$ (2026). VISTA: Velocity-Informed Smart Transmission Adaptation in High-Mobility IoT. In *Proc. International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks Workshops (WiOpt Workshops 2026)*, pp. 9–15, Jun. 2026.

[C4] Busacca, F., **Panebianco, A.**, & Sun, Y.$^{\alpha}$ (2025). Adaptive Underwater Acoustic Communications with Limited Feedback: An AoI-Aware Hierarchical Bandit Approach. In *Proc. IEEE Global Communications Conference (GLOBECOM 2025)*, pp. 2426–2431, Dec. 2025.

[C5] Busacca, F., Galluccio, L., **Panebianco, A.**, Palazzo, S., Raftopoulos, R., & Scuderi, D. (2025). MERMAID: A Multi-Armed Bandit Approach for Optimal Carrier Selection in OFDM-Based Underwater Acoustic Networks. In *Proc. IEEE International Symposium on Personal, Indoor and Mobile Radio Communications Workshops (PIMRC Workshops 2025)*, pp. 1–6, Sep. 2025.

[C6] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R.$^{\alpha}$ (2025). POSEIDON: A Multi-Armed Bandit Framework for Power and Modulation Scheme Adaptation in Underwater Acoustic Networks. In *Proc. IEEE International Conference on Communications Workshops (ICC Workshops 2025)*, pp. 2045–2050, Jun. 2025.

[C7] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R.$^{\alpha}$ (2025). A Distributed Multi-Armed Bandit Approach for Modulation Adaptation in Underwater Networks. In *Proc. IEEE Wireless Communications and Networking Conference (WCNC 2025)*, pp. 1–6, Mar. 2025.

[C8] Scarvaglieri, A., **Panebianco, A.**, & Busacca, F. (2024). MAGELLAN: A Distributed MAB-Based Algorithm for Energy-Fair and Reliable Routing in Multi-Hop LoRa Networks. In *Proc. IEEE Global Communications Conference (GLOBECOM 2024)*, pp. 2250–2255, Dec. 2024.

[C9] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R.$^{\alpha}$ (2024). Adaptive Modulation in Underwater Acoustic Networks (AMUSE): A Multi-Armed Bandit Approach. In *Proc. IEEE International Conference on Communications (ICC 2024)*, pp. 2336–2341, Jun. 2024. *Selected Best Paper.*

[C10] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Scarvaglieri, A.$^{\alpha}$ (2024). Balancing Optimization for Underwater Network Cost Effectiveness (BOUNCE): A Multi-Armed Bandit Solution. In *Proc. IEEE International Conference on Communications Workshops (ICC Workshops 2024)*, pp. 1340–1345, Jun. 2024.

[C11] Busacca, F., Galluccio, L., Palazzo, S., & **Panebianco, A.**$^{\alpha}$ (2023). Underwater Acoustic Channel Models for SNR Prediction in a Real Shallow-Water Environment. In *Proc. IEEE International Conference on Communications Workshops (ICC Workshops 2023)*, pp. 721–726, May 2023. *Based on experimental acoustic traces provided by NATO CMRE.*

Posters and Demonstrations

[P1] Chakraborty, S.†, **Panebianco, A.**†, Tian, Y., Chan, K. S., Dagefu, F., Sun, Y., & Shroff, N. B. (2026). Distributed Cross-Layer Optimization for Covert Multi-Hop, Multi-Modal Networks: Exponentially Fast Convergence and Robust Tracking. Poster, *IEEE Military Communications Conference (MILCOM 2026), Young Scholar Workshop*, Washington, DC, USA, Oct. 2026.

[P2] McDowell, C.†, **Panebianco, A.**†, & Sun, Y. (2026). MetaVIB-Air: An SDR Testbed for Semantic Pedestrian Safety Monitoring. Demonstration, *IEEE Military Communications Conference (MILCOM 2026), Workshop on Quality, Value, and Age of Information for Tactical Networks (WS7)*, Washington, DC, USA, Oct. 2026.

[P3] **Panebianco, A.** (2023). Characterization of Underwater Acoustic Channel Models for SNR Prediction in Real Shallow Water Environment. Poster, *Annual Meeting of the Italian National Group on Telecommunications and Information Theory (GTTI 2023)*, Rome, Italy, Sep. 2023. *Based on experimental acoustic traces provided by NATO CMRE. Francesco Carassa Award (Best Poster).*

Manuscripts Under Review

[U1] **Panebianco, A.**, Busacca, F., & Sun, Y. (2026). AoI-Regulated Underwater Acoustic Link Adaptation: A Bilevel Bandit Framework with Decision-Dependent Delay. Submitted to *IEEE Transactions on Communications*; under review. *Extended version of [C4].*

[U2] Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, Raftopoulos, R., & Scuderi, D.$^{\alpha}$ (2026). A Fully Distributed Framework for Subcarrier and Power Allocation in Underwater OFDM Networks: Exploiting Game Theory to Support Online Learning. Submitted to *IEEE Transactions on Communications*; under review. *Extended version of [C5].*

[U3] Busacca, F., Palazzo, S., **Panebianco, A.**, Raftopoulos, R., & Scarvaglieri, A.$^{\alpha}$ (2026). Exploiting Multi-Armed Bandit in Wireless Networks: A Survey on Lightweight Decision-Making for Resource-Constrained Devices. Submitted to *IEEE Communications Surveys & Tutorials*.

PATENT APPLICATIONS

[PA1] Sun, Y., Chakraborty, S., McDowell, C. D., & **Panebianco, A.** (2026). Significance-Driven Semantic Communication. U.S. Provisional Patent Application No. 64/065,591, filed 14 May 2026.

[PA2] Sun, Y., Chakraborty, S., Chamoun, S. G., McDowell, C. D., **Panebianco, A.**, & Ross, T. P. (2025). Spectrum Efficiency of Semantics and Goal Oriented Communications. U.S. Provisional Patent Application No. 63/875,119, filed 3 Sep. 2025.

RESEARCH PROJECTS

Participation as a researcher; sponsor and Principal Investigator (PI) are indicated where documented.

HEIDRA – Heterogeneous Edge Intelligence for Natural Disaster Response Analytics Sep. 2026 – Present

Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA

Role: postdoctoral researcher

Sponsor: U.S. National Science Foundation, Verticals-Enabling Intelligent Network Systems (VINES) Track 2 collaborative award. PI: Christopher G. Brinton (Purdue University). Co-PIs include Alexei Ashikhmin (Nokia Bell Labs), Shreyas Sundaram (Purdue University), Christopher Vo (Saab, Inc.), and Yin Sun (Auburn University). Partners: Purdue University, Auburn University, University of Virginia, Saab, Nokia.

Resilient edge-intelligence platform for natural-disaster response that integrates mobile and multi-access edge computing with NextG wireless technologies to maintain reliable connectivity, enable rapid resource adaptation and reallocation, and provide on-site and on-device intelligence to emergency teams and first responders. The Auburn contribution develops importance-based, task-oriented networking that prioritizes mission-critical information over power- and capacity-limited disaster-zone networks.

Development of NextG Wireless Testbeds for Semantic and Goal-Oriented Communication Jan. 2026 – Present

Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA

Role: postdoctoral researcher

Sponsors: Alabama Research and Development Enhancement Fund (ARDEF), grant agreement 1ARDEF26 11; NVIDIA Academic Grant Program. PI: Yin Sun (Auburn University).

Development and integration of end-to-end NextG testbeds for semantic and goal-oriented communication, with emphasis on protocol implementation, experimental validation, resource-constrained networking, and mission-critical IoT applications. The testbeds include SafeStep@AU (safestep.eng.auburn.edu), a live semantic-communication testbed for pedestrian safety monitoring at a signalized intersection in Auburn, AL. Outputs: [C2, P2].

Distributed Network Control in Dynamic Military Environments: A Learning-Based Approach — From Theory to Implementation Jan. 2026 – Sep. 2026

Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA

Role: postdoctoral researcher

Sponsor: U.S. Army Research Office (ARO), cooperative agreement W911NF-24-2-0205. PI: Yin Sun (Auburn University). Co-PIs: Ness B. Shroff (The Ohio State University), Kaushik R. Chowdhury (The University of Texas at Austin).

Learning-based distributed network control for dynamic environments, including distributed cross-layer optimization for covert multi-hop, multi-modal networks, developed with The Ohio State University and the DEVCOM Army Research Laboratory. Outputs: [C1, P1].

AQUASMARTT – Data Acquisition with Underwater and Above-Water Sustainable Wireless Nodes for Monitoring How Climatic Changes Affects Natural Disasters and Biodiversity Nov. 2022 – Dec. 2025

Department of Engineering, University of Palermo, Palermo, Italy

Role: Ph.D. researcher

Sponsor: European Union – NextGenerationEU, Italian National Recovery and Resilience Plan (PNRR), Italian Ministry of University and Research (MUR) partnership PE00000001 "RESTART – Telecommunications of the Future", Spoke 6 "Innovative Architectures and Extreme Environments", focused project F16. PI: Michele Zorzi (University of Padova). Partners: University of Padova, University of Catania, Sapienza University of Rome.

Research and development of intelligent networking components for coastal sensing and underwater acoustic networks: channel-quality prediction, adaptive modulation and power control, AoI-aware feedback scheduling, energy-efficient resource allocation, DESERT-based simulation and protocol-level evaluation for IoUT scenarios, and integration aspects of heterogeneous underwater sensing and communication systems. Outputs: [J4] (listed among the project's publications); [J2, C4] (acknowledging the RESTART programme).

4Frailty – Smart Sensors, Infrastructures, and Management Models for the Safety of Frail Patients Jul. 2023 – Oct. 2023

Department of Engineering, University of Palermo, Palermo, Italy

Role: Ph.D. researcher

Sponsor: MUR, National Operational Programme (PON) "Ricerca e Innovazione" 2014–2020 and Development and Cohesion Fund (FSC), project ARS01_00345 ("Technologies for Living Environments"). PI: Bruno Andò (University of Catania). Partners: 21 organizations, led by Distretto Tecnologico Sicilia Micro e Nano Sistemi.

IoT-based remote-care monitoring platform integrating wearable sensors, Android interfaces, MQTT-based data exchange, LoRa connectivity, and secure data storage.

Secure and Efficient Wireless Communications – Intelligent Communication Algorithms and Integrated System Development for Underwater Wireless Communication Environments Sep. 2020 – May 2021
Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
Role: research intern
PI: Alfio Lombardo (University of Catania)
Design, integration, and field testing of intelligent communication and monitoring architectures for underwater wireless environments: nearshore shallow-water marine trials of an integrated acoustic/LoRa IoUT system at La Playa, Catania (underwater acoustic modems, sensing devices, control boards, LoRa connectivity, remote monitoring interfaces; evaluation of transmission delay, packet delivery, and achievable bit rate for multimedia and control data); IoT-based sensing, MQTT-based data exchange, digital-twin modules, web dashboards, integrated sensing/communication prototypes, and simulated wake-up radio mechanisms. Outputs: [J6].

PROFESSIONAL SERVICE AND RESEARCH MANAGEMENT

Research Management

Research Proposal and Grant Preparation Jan. 2026 – Present
Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA
Contributing author of research proposals with the project supervisor: technical sections, problem formulation, research planning, and alignment with project and sponsor objectives.

Conference Organization

Organizing Committee Member Nov. 2026
3rd International Workshop on the Integration between Distributed Machine Learning and the Internet of Things (AIoT 2026), in conjunction with the 27th ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc 2026), Tokyo, Japan

Finance and Registration Chair Jun. 2026
24th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt 2026), Columbus, OH, USA

Technical Program Co-Chair Jun. 2026
3rd Workshop on Modeling and Optimization for Semantic Communications (MOSC 2026), in conjunction with WiOpt 2026, Columbus, OH, USA

Technical Program Committees

Technical Program Committee Member Oct. 2026
IEEE Military Communications Conference (MILCOM 2026), Track 1 (Waveforms and Signal Processing) and Workshop on Quality, Value, and Age of Information for Tactical Networks (WS7), Washington, DC, USA

Peer Review

Journal Reviewer 2024 – Present
IEEE/ACM Transactions on Networking; IEEE Transactions on Network Science and Engineering; IEEE Transactions on Vehicular Technology; IEEE Internet of Things Journal; IEEE Vehicular Technology Magazine; Nano Communication Networks; Entropy

INVITED TALKS AND SEMINARS

A Multi-Armed Bandit Approach in Underwater Networks Feb. 2024
Winter School on Underwater Network Simulations and Experimentation (UNWiS 2024), Department of Information Engineering, University of Padova, Padova, Italy

Design and Experimentation of an Integrated Acoustic/LoRa System for Transmission of Multimedia Sensor Data over an Internet of Underwater Things May 2021
Seminar on Development and Applications of Innovative Materials and Processes for the Diagnostics and Restoration of Cultural Heritage, Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
Related publication: [J6].

TEACHING EXPERIENCE

Teaching Assistant Mar.–Jun. 2023
Mar.–Jun. 2024
Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
Course: "IoT and Big Data Sensing, Compression, and Communication"
Support for teaching activities, laboratory sessions, and student supervision on IoT systems, sensing architectures, data compression, and communication technologies, with a focus on practical implementation and performance evaluation.

MENTORING AND SUPERVISION

Ph.D. Research Mentor Jan. 2026 – Present
Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA
Students: Sirin Chakraborty (co-first author of [C1, P1]; co-author of [C2]; co-inventor of [PA1]), Christian McDowell (co-first author of [C2, P2]; co-inventor of [PA1]), Hutama Bramantyo

Undergraduate Research Mentor May 2026 – Aug. 2026
Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA
Student: Jeremiah Yang (co-first author of [C2])

M.Sc. Thesis Co-Supervisor Jan. 2025 – Oct. 2025
Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
Student: Dario Scuderi (co-author of [C5, U2])
Thesis: "Smart Carrier Selection in OFDM-Based Acoustic Networks for the Internet of Underwater Things" (M.Sc. in Communications Engineering)

TECHNICAL SKILLS

- **Learning and optimization:** multi-armed bandits (stochastic, contextual, combinatorial, hierarchical, restless, multi-player/fully distributed); online learning; reinforcement and deep reinforcement learning; distributed convex optimization (ADMM); VIB-based semantic encoding; AoI-aware scheduling and optimization.
- **Channel modelling and signal processing:** finite-state Markov and hidden Markov models; Kalman filtering; data-driven channel prediction; underwater acoustic trace processing; channel impulse response analysis; SNR estimation; channel-quality characterization; propagation-aware link modelling; physical-layer performance analysis.
- **Wireless networking:** semantic and goal-oriented communication; cross-layer resource allocation; link adaptation (modulation, power, OFDM carrier selection); feedback scheduling; routing and protocol design; energy-aware, covert, and jamming-resilient networking; underwater acoustic networks and IoUT; LoRa and MQTT-based IoT.
- **Experimental systems:** SDR and NextG testbeds; computer vision pipelines for camera-based pedestrian monitoring; underwater acoustic modem deployment and field trials; acoustic/LoRa system integration; edge-computing platforms; experimental data acquisition and analysis.
- **Software and tools:** Python, MATLAB, Java, and Arduino; DESERT Underwater network simulator, Mininet, and Wireshark; algorithm prototyping, network simulation, and protocol-level evaluation; web-based demonstrators and dashboards; LaTeX.
- **Research and communication:** scientific writing; technical reporting and presentation; research proposal preparation; problem formulation and research planning; literature review.

QUALIFICATIONS, MEMBERSHIPS, AND PERSONAL INFORMATION

Professional Engineer Licence (Information Engineering Sector) Nov. 2023
Italian State Examination (Esame di Stato), University of Catania, Catania, Italy

Certificate in ESOL International (CEFR Level C2) May 2022
British Institutes

Memberships: IEEE; IEEE Communications Society; IEEE Young Professionals

Languages: Italian (native); English (CEFR C2)

Nationality: Italian